DentaMind AI

Dental X-ray analysis research prototype

A complete guide for the project team: what it does, how it works, what the models achieve, what we tested, and what is still open.



Every output is a research draft: a person must review it. Nothing here is a diagnosis.

Version of 15 October 2026. All numbers are copied from the project's own report files (named on each page). Validation numbers are optimistic: the same split chose each model's best training epoch.

Research use only. DentaMind AI is not a medical device, is not clinically validated, and must not guide treatment. Every output is an unconfirmed research draft until a named person reviews it.

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1. The project in one page

DentaMind AI looks at a **panoramic dental X-ray** (an OPG, the wide image that shows all teeth at once) and produces a structured, reviewable draft: where each tooth is, its tooth number, its outline, which of four findings it most resembles, how unsure the model is, and a draft report in English, Hindi or Marathi.

What it can do

- **Find teeth** on the X-ray and draw a box and an outline around each one.
- **Suggest a tooth number** in the international FDI system (11-48). Unsure numbers are marked "?".
- **Name the closest trained finding** for each tooth: Caries, Deep Caries, Impacted or Periapical Lesion.
- **Show doubt**: uncertainty scores and image-quality warnings, with a list of regions a person should check.
- **Write a draft report** from the structured results, without any external service.
- **Support research work**: model registry, evaluation reports, active learning, reviewer workflow and audit trails.

What it cannot do

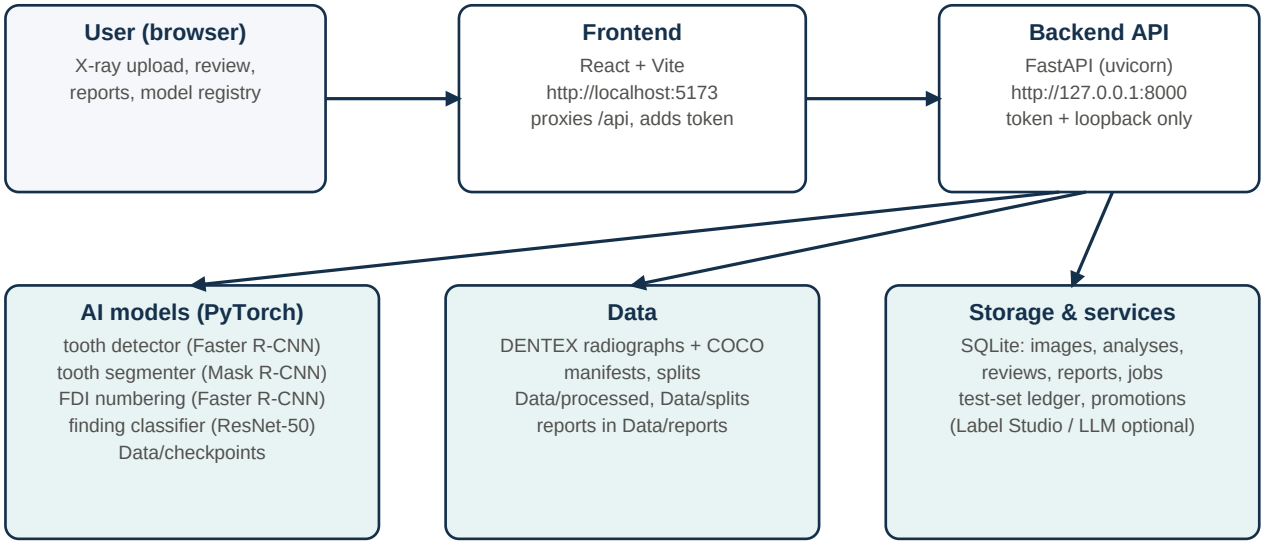
- **It cannot say a tooth is healthy.** The classifier only knows four findings, so it always picks one of them.
- It is **not a diagnosis** and has **not been clinically validated**. It was trained on one public dataset (DENTEX).
- It cannot prove that two different X-rays of the same patient were kept apart, because DENTEX has no patient IDs.

Key numbers at a glance

Question	Answer (source: Data/reports)
How well are teeth found?	Recall 0.98-0.996, precision about 0.96 (validation); detection is reliable
How good are tooth outlines?	Mask AP@0.5 0.97-0.98; Dice 0.87 on the test split
How accurate are tooth numbers?	93.6 % (validation) / 96.8 % (test) of matched teeth exactly right
How accurate is the finding classifier?	Accuracy 0.79 and macro-F1 0.63 end-to-end in the app (validation); strong for Caries/Impacted, weak for Deep Caries and Periapical Lesion
Is the data split clean?	17 of 18 leakage checks pass; 1 warning that cannot affect any metric

2. How the system is built

Three parts run on one computer. The browser talks only to the frontend. The frontend forwards requests to the backend API and adds a secret token on the server side, so the token never reaches the browser. The backend loads the AI models, reads the data and stores results in a local SQLite database.



Part	Technology	Where	Job
Frontend	React 18, Vite, Tailwind	frontend/	Pages: X-ray analysis, review, reports, model registry, about & limitations
Backend	Python 3.13, FastAPI, uvicorn	backend/	About 70 API routes; security (token, loopback-only, rate limit)
Core library	Python, PyTorch 2.6, torchvision	src/dentexmodel/	Data handling, training, inference, evaluation, reports
Models	Faster/Mask R-CNN, ResNet-50	Data/checkpoints/	4 trained checkpoints + metadata sidecars
Database	SQLite	runtime/dentamind.sqlite3	Images, analyses, reviews, reports, jobs, promotions
Scripts	Python	scripts/	Training, evaluation, audits, experiments

Security in short: the API answers only on 127.0.0.1, every route except the health check needs the token (*runtime/api_token.txt*), and requests are rate-limited. There is no user login, so the app must stay on this computer.

3. How an X-ray is analysed



Every output is a research draft: a person must review it. Nothing here is a diagnosis.

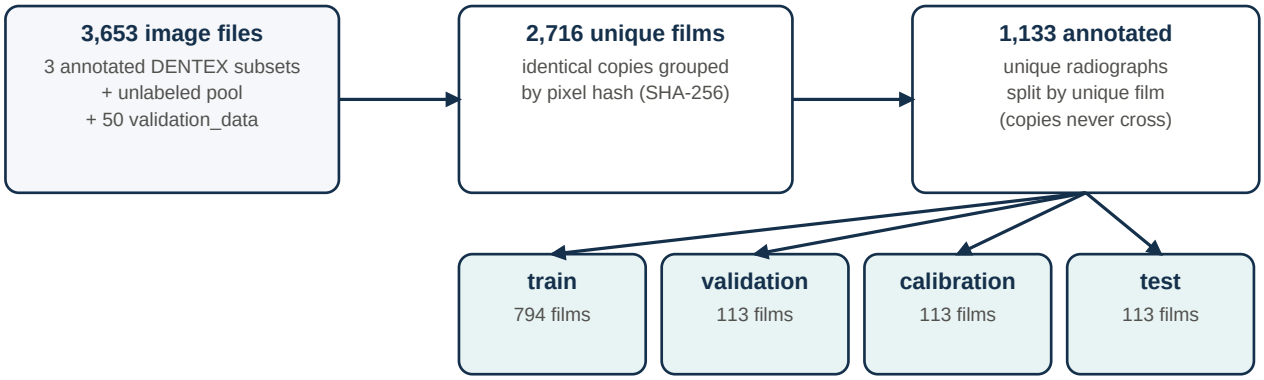
- **1 Load.** The image is read as grayscale and copied into three identical channels. It is not contrast-adjusted for the models.
- **2 Detect.** A Faster R-CNN finds every tooth. Boxes with a score below 0.3 are dropped.
- **3 Number.** A second Faster R-CNN predicts FDI tooth numbers; a number is attached to a box only if the two overlap well (IoU at least 0.5). Teeth without a match show "R7"-style region names instead of a number.
- **4 Segment.** A Mask R-CNN draws an outline for each tooth.
- **5 Classify.** Each detected box is cut out with 10 % padding, resized to 224 x 224 pixels, and a ResNet-50 names the *closest* of the four findings. This is shown with a "~" (for example "~Caries") because it is not a confirmed finding.
- **6 Doubt.** Entropy of the classifier and disagreement on a flipped copy flag teeth for review. Thresholds were chosen on the calibration split and are labelled provisional.
- **7 Report.** A template turns the structured results into a draft report in English, Hindi or Marathi. Hindi and Marathi wording has not been reviewed by a native-speaking dentist.

Colours on the X-ray

Colour	Label	Meaning
Teal	14	Tooth number from the numbering model, consistent with its neighbours
Red	14?	Proposed number that needs checking (red means "check the number", not disease)
Grey	R7	Region without a tooth number; 7 is only the detection order
Orange	-	The region you selected

4. The data

The project uses **DENTEX**, a public dataset of panoramic X-rays. It comes as three annotated subsets with different labels, plus unlabeled images. The subsets overlap: the same X-ray often appears in two or three of them under different file names. We therefore count and split **unique radiographs** (identical pixels = same film), not files.

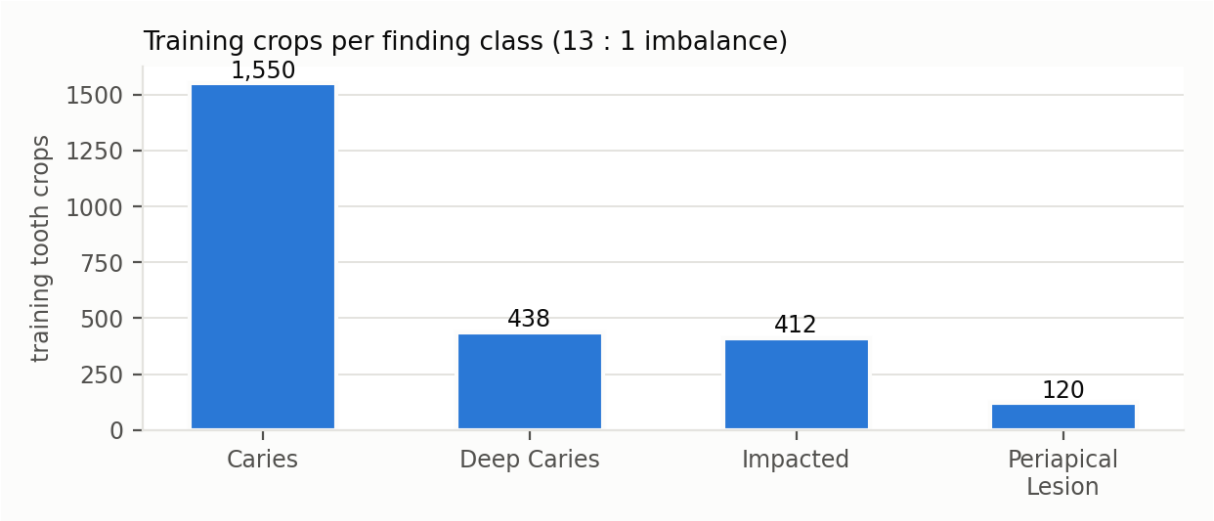


Subset	What is labelled	Used to train
quadrant	4 large quadrant regions per image (not teeth)	not used as tooth ground truth
quadrant_enumeration	every tooth: box, outline, FDI number	detector, segmenter, numbering model
quadrant_enumeration_disease	only diseased teeth, each with one of 4 findings	finding classifier
unlabeled pool (1,598 files)	nothing	active-learning pool only
validation_data (50 files)	nothing	demo images only; never scored

Splits

Train / validation / calibration / test are assigned per unique radiograph: **794 / 113 / 113 / 113**. Training uses **train**; **validation** chooses the best training epoch; **calibration** sets uncertainty thresholds; **test** is the final check (already used once, see section 9).

The classifier's imbalance



Caries has 13 times more examples than Periapical Lesion. The validation split has only 9 Periapical Lesion and 45 Deep Caries crops, so their scores move a lot with each tooth. Source: classifier sidecar *class_counts_train*.

5. The four AI models

Model	Architecture	Trained on	Output
dentex_teeth_detection_v1	Faster R-CNN ResNet-50 FPN (COCO start)	440 train radiographs, all teeth	one box per tooth
dentex_teeth_segmentation_v1	Mask R-CNN ResNet-50 FPN (COCO start)	440 train radiographs, tooth outlines	outline per tooth
dentex_fdi_numbering_v1	Faster R-CNN, 32 FDI classes	440 train radiographs	tooth number per box
dentex_findings_classification_v1	ResNet-50 (ImageNet start)	2,520 tooth crops, 476 radiographs	closest of 4 findings

Each checkpoint has a metadata file (sidecar) with its task, classes, training settings, data version and a SHA-256 hash. The app refuses to load a model whose file no longer matches its hash. All four are **legacy** for provenance: they were trained before we started recording exactly which images each model saw (section 9).

Training settings (classifier)

ResNet-50, 15 epochs, batch 32, AdamW learning rate 3e-4, cosine schedule, inverse-frequency class weights, horizontal flip and mild contrast jitter, best epoch chosen by validation macro-F1 (epoch 5 for the deployed model). Re-running the production trainer today reproduced the deployed weights **bit for bit**.

6. How good are the models?

Validation = the split used to pick the best epoch (slightly optimistic). Test = historical final check on 2026-10-14. Sources: *Data/reports/eval_*_val.json* and *eval_*_test.json*.

Teeth: detection, outlines and numbers

Model / metric	Validation	Test
Detection precision / recall (score 0.3, IoU 0.5)	0.962 / 0.984	0.960 / 0.974
Detection box mAP / AP50	0.560 / 0.978	0.515 / 0.966
Segmentation precision / recall (box level)	0.966 / 0.986	0.962 / 0.980
Segmentation mask AP@0.5	0.980	0.974
Segmentation Dice / IoU (matched teeth)	not recorded	0.873 / 0.780
Numbering: exact FDI number among matched teeth	93.6 %	96.8 %
Numbering: correct quadrant	99.7 %	99.9 %

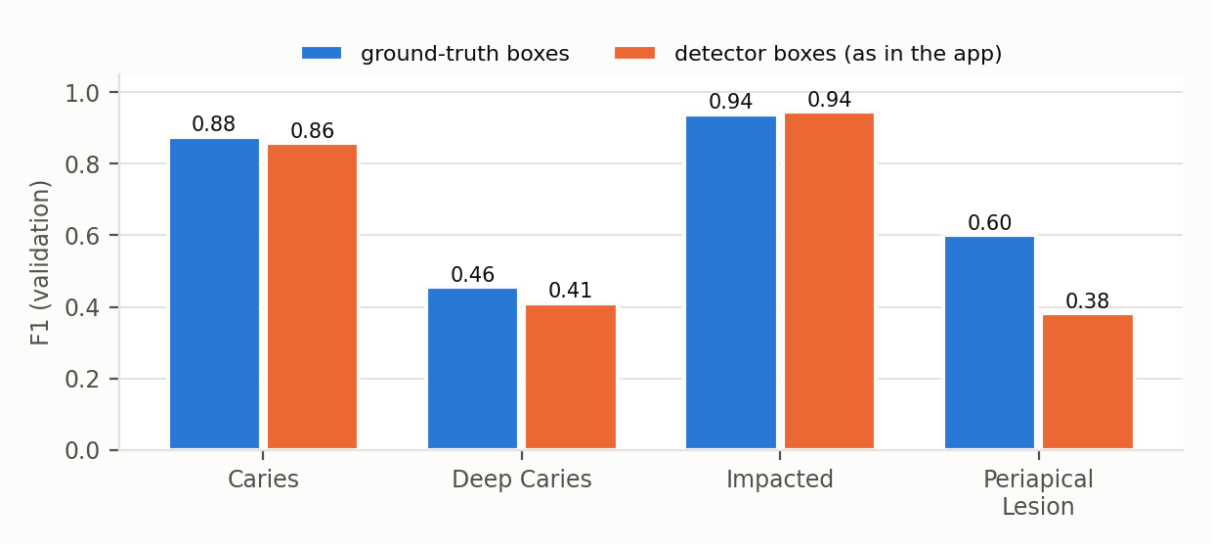
Findings: the classifier (ground-truth tooth boxes)

Metric	Validation (331 crops)	Test (354 crops)
Accuracy	0.825	0.785
Macro precision / recall / F1	0.693 / 0.763 / 0.718	0.653 / 0.649 / 0.646
Caries P / R / F1	0.885 / 0.863 / 0.874	0.866 / 0.820 / 0.842
Deep Caries P / R / F1	0.500 / 0.467 / 0.483	0.414 / 0.500 / 0.453
Impacted P / R / F1	0.919 / 0.944 / 0.932	0.887 / 0.969 / 0.926
Periapical Lesion P / R / F1	0.467 / 0.778 / 0.583	0.444 / 0.308 / 0.364

Reading this: teeth are found, outlined and numbered well. The classifier is good for Caries and Impacted and weak for the two rare findings. On the test split it found only 4 of 13 periapical lesions.

7. Realistic performance in the app

The numbers above use the *true* tooth boxes. In the app the classifier receives the *detector's* boxes, which are slightly different. We measured both on the same 323 validation teeth (source: *end_to_end_validation_evaluation.md*).



Metric (validation)	True boxes	Detector boxes	End-to-end (missed tooth = wrong)
Accuracy	0.830	0.811	0.792
Macro-F1	0.717	0.649	0.634
Deep Caries recall	0.439	0.366	0.333
Periapical Lesion recall	0.857 (6/7)	0.571 (4/7)	0.444 (4/9)

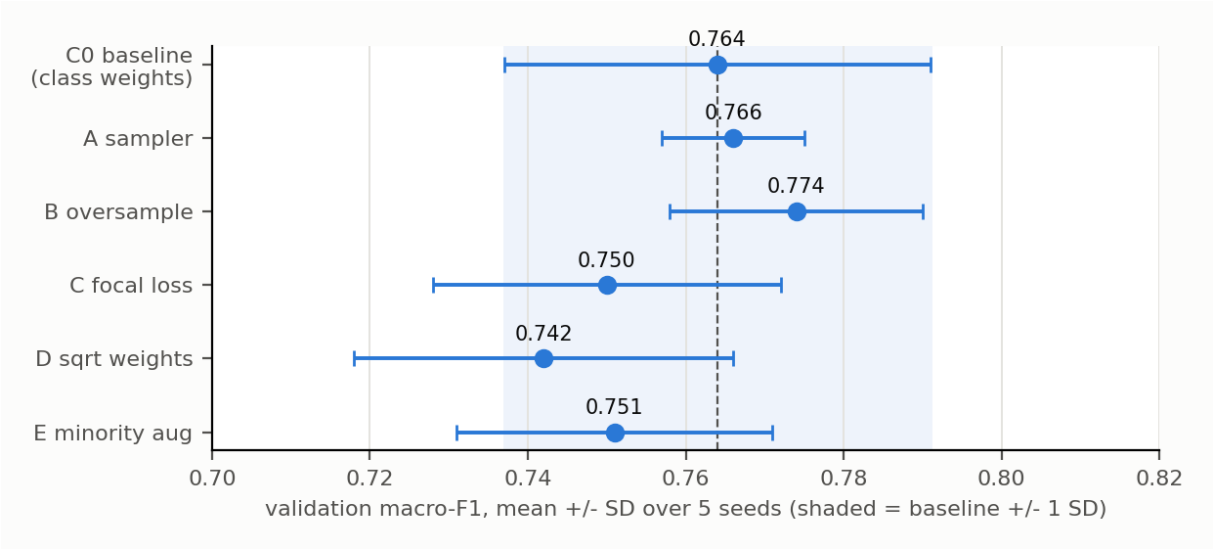
Why the gap

- **Small box shifts change predictions:** about 10 % of teeth get a different answer even when the detector box overlaps the true box by more than 90 %.
- **Some errors exist even with perfect boxes** (41 of 331 findings).
- **Two findings on one tooth:** 8 findings (including 4 of the 9 periapical lesions) sit on teeth that also carry another finding. A one-answer-per-tooth classifier can never get both right.
- **Missed teeth are rare:** only 4 findings were missed by the detector.
- **Most teeth have no recorded finding but still get a class** (1,580 of 1,903 detected teeth). They are not proven healthy, so we cannot count them as right or wrong, but users should know the app labels every tooth.

8. Experiments we ran to improve the classifier

Every experiment used the same data, 5 random seeds (42-46) and rules written **before** seeing results: a method counts as better only if it beats the baseline in at least 4 of 5 seeds by more than 0.027 macro-F1 without hurting the rare classes.

A. Class-imbalance methods (validation, true boxes)



None beat the baseline. Sampling and oversampling raised precision but lowered rare-class recall; focal loss and square-root weights were worse; minority-only augmentation cut Deep Caries recall in every seed. Tuning decision thresholds on validation looked better in-sample (+0.023) but was worse when checked properly (-0.005). Source: *docs/audit_evidence/ml_improvement/imbalance/report.md*.

B. Crop-robustness (training on jittered boxes)

To make the classifier tolerant of detector boxes, we trained on boxes jittered the way the detector really errs (measured on training images only). Detector-box macro-F1: **0.720 +/- 0.038** vs baseline **0.723 +/- 0.016**, better in only 1 of 5 seeds: **did not pass**. It raised rare-class recall but lowered their precision. Source: *crop_robustness_experiment.md*.

C. Production recipe vs experiment recipe

The deployed classifier scored below every run of the experiment recipe on detector boxes, so we ran a pre-registered comparison. Re-running the production trainer reproduced the deployed model **bit for bit**. Across 5 seeds the two recipes are equal: detector-box macro-F1 **0.722 +/- 0.045** (production) vs **0.724 +/- 0.015** (experiment); the experiment recipe won only 10 of 25 run pairs and did not pass. **The deployed model is simply the weakest of five runs** of its recipe (0.651; the others 0.718-0.776). Picking the best run by its validation score would be optimistic, so any replacement needs its own pre-registered selection and evaluation. Source: *training_recipe_comparison_results.md*.

9. Research integrity: how we keep results honest

Protection	What it does	Enforced in code?
Duplicate-aware splits	Identical films always land in the same split (checked by pixel hash)	Yes
Leakage audit (18 checks)	17 pass; 1 warning: 3 unlabeled demo files equal test films (never scored)	Yes, re-runnable
Test-set ledger	Every test evaluation needs a named person and a reason, is logged with hashes, one per model; the ledger is hash-chained so edits are detectable	Yes (files can still be edited)
Training manifests	New models record exactly which images they were trained on	Yes, for new models
Model promotion	Needs a validation report, a named approver and a written reason; rollback recorded	Yes
Uploads vs reviews	An upload identical to a labelled film can never feed training	Yes
Patient separation	Not possible: DENTEX has no patient IDs	No - open limitation

The internal test split has already been used for all four current models (2026-10-14). Their test numbers are historical. An honest estimate for a future model needs a new, external test set (protocol in docs/RESEARCH_INTEGRITY.md).

10. Known limitations

- No "healthy" class: every tooth is given one of four findings.
- Deep Caries and Periapical Lesion are weak (end-to-end recall about 0.33 and 0.44 on validation).
- Few rare-class examples: 9 Periapical Lesion and 45 Deep Caries validation crops; results for them are noisy.
- A tooth can carry two findings, but the classifier gives one answer per tooth.
- One public dataset; no external test; behaviour on other scanners is unknown.
- Patient-level separation cannot be proven.
- The current models were trained before image-level training records existed (legacy provenance).
- Hindi/Marathi reports not reviewed by a native-speaking dentist; reviewer names are typed in, not verified.
- No user accounts: the app must run only on this computer.

11. What to do next

- **Decide how to replace the weak deployed run** (section 8C): a pre-registered rule for choosing among several runs, evaluated without reusing the validation numbers used to choose.
- **Collect more dentist-labelled Deep Caries and Periapical Lesion examples** - the step most likely to help.
- **Add a reliable "no finding" class**, which needs labels confirming that teeth have no finding.
- **Switch to a multi-label classifier** so a tooth can have two findings (for example Caries + Periapical Lesion).
- **Obtain an external test set** and evaluate one final model once.

12. Running the project

Open two terminals in **D:\Dentex-Gen Ai**:

What	Command	Then open
Backend API	.venv\Scripts\python.exe -m uvicorn backend.main:app --host 127.0.0.1 --port 8000	http://127.0.0.1:8000/api/health
Frontend	cd frontend then npm.cmd run dev	http://localhost:5173/

- API documentation: http://127.0.0.1:8000/docs (needs the token from runtime\api_token.txt).
- Run all unit tests: *.venv\Scripts\python.exe -m unittest discover -s tests* (233 tests).
- Re-run the leakage audit: *.venv\Scripts\python.exe scripts\check_leakage.py*.
- Check the test-set ledger: *.venv\Scripts\python.exe scripts\test_set_ledger.py --verify*.
- GPU note: the laptop GPU has 4 GB. Run one GPU job at a time; restart the backend before training so it releases GPU memory.

13. Glossary

Term	Meaning
OPG / panoramic X-ray	One wide X-ray showing all teeth and both jaws
FDI number	International tooth numbering: first digit = quadrant (1-4), second = tooth (1-8), e.g. 36
Bounding box	A rectangle around a tooth
IoU	Overlap of two boxes: shared area / combined area (1 = identical)
Precision	Of the items the model flagged, how many were right
Recall	Of the real items, how many the model found
F1	One number combining precision and recall (their harmonic mean)
Macro-F1	F1 averaged over the 4 classes equally, so rare classes count as much as common ones
Balanced accuracy	Average recall over the classes
AP / mAP	Average precision over score thresholds (detection quality)
Dice	Overlap of two outlines (1 = identical); similar to IoU
Train / validation / calibration / test	Data for learning / choosing the epoch / setting thresholds / final check
Seed	Starting value of the random generator; 5 seeds = 5 repeats to measure luck
Data leakage	The same image (or patient) ending up in training and testing, which inflates results
Checkpoint / sidecar	A saved trained model file / its metadata file

14. Where to find things

File or folder	What is in it
README.md	Project overview and setup
docs/KNOWN_LIMITATIONS.md	Full list of limitations
docs/RESEARCH_INTEGRITY.md	Test-set use, provenance, promotion: what is enforced vs procedural
docs/MODEL_TRAINING.md	How each model was trained
Data/reports/eval_*.json / .md	Evaluation of each model on validation and test
Data/reports/end_to_end_validation_evaluation.md	Detector -> classifier performance as in the app
Data/reports/crop_robustness_experiment.md	Crop-jitter experiment
Data/reports/training_recipe_comparison_results.md	Recipe comparison: protocol and results
Data/reports/leakage_check.md	Leakage audit results
Data/reports/test_set_ledger.json	Record of every test-split use
docs/audit_evidence/ml_improvement/	All experiment runs, per seed
documents/build_project_guide.py	Rebuilds this PDF

Questions about a number in this guide: open the source file named next to it - every figure here was copied from those files, not estimated.